

TA-CNN: A Unified Network for Human Behavior Analysis in Multi-Person Conversations

Fuyan Ma
Hunan University
Changsha, China
mafuyan@hnu.edu.cn

Bin Sun*
Hunan University
Changsha, China
sunbin611@hnu.edu.cn

Ziyu Ma
Hunan University
Changsha, China
maziyu@hnu.edu.cn

Shutao Li
Hunan University
Changsha, China
shutao_li@hnu.edu.cn

ABSTRACT

Human behavior analysis in multi-person conversations has been one of the most important research issues for natural human-robot interaction. However, previous datasets and studies mainly focus on single-person behavior analysis, therefore, can hardly be generalized in real-world application scenarios. Fortunately, the MULTIMEDIATE'22 Challenge provides various video clips of multi-party conversations. In this paper, we present a unified network named TA-CNN for both sub-challenges. Our TA-CNN can not only model the spatio-temporal dependencies for eye contact detection, but also capture the group-level discriminative features for multi-label next speaker prediction. We empirically evaluate the performance of our method on the officially provided datasets. Our method achieves the state-of-the-art result of 0.7261 for eye contact detection in terms of accuracy and the UAR of 0.5965 for next speaker prediction on the corresponding test sets.

CCS CONCEPTS

• **Computing methodologies** → **Activity recognition and understanding**; *Supervised learning*; **Neural networks**; • **Human-centered computing** → HCI theory, concepts and models.

KEYWORDS

Deep learning, eye contact detection, next speaker prediction, multi-person conversation

ACM Reference Format:

Fuyan Ma, Ziyu Ma, Bin Sun, and Shutao Li. 2022. TA-CNN: A Unified Network for Human Behavior Analysis in Multi-Person Conversations. In *Proceedings of the 30th ACM International Conference on Multimedia (MM '22)*, October 10–14, 2022, Lisboa, Portugal. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3503161.3551587>

*Bin Sun is the corresponding author.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than ACM must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from [permissions@acm.org](https://permissions.acm.org).

MM '22, October 10–14, 2022, Lisboa, Portugal

© 2022 Association for Computing Machinery.

ACM ISBN 978-1-4503-9203-7/22/10...\$15.00

<https://doi.org/10.1145/3503161.3551587>



Figure 1: Video samples from the MPIIGroupInteraction dataset [20]. We detect and align these facial regions by OpenFace [1] to filter out the redundant background information.

1 INTRODUCTION

Understanding human behaviors in multi-party conversations is among the most fundamental tasks to achieve natural and effective human-robot interaction. Human can interact not only by the verbal behavior (i.e., speech), but also by non-verbal behavioral cues (e.g. eye contact). To naturally participate in human group conversations, a robot need have the ability of *eye contact detection* and *next speaker prediction*, which are also the two sub-challenges of MULTIMEDIATE [18, 19].

How to perform human behavior analysis in real-world multiparty settings has been an emerging topic, attracting more and more researchers focus on it. It is an interdisciplinary research field, spanning from computer science to sociolinguistics. Specifically, eye contact in non-verbal human communication is one of the most crucial ways to attract and acknowledge attention from others. Quantification of eye contact in multi-person conversation is valuable as a part of the analysis of social relationships between participants and the expression of attentiveness. In addition, next speaker prediction (also known as turn-changing prediction) is especially important for fully understanding the mechanisms of human face-to-face communication. Accurately predicting who will speak next in group meetings can avoid utterance collisions and achieve smooth communication for human-robot interaction. Human behavior analysis in multi-party conversations also has numerous medical or psychiatric applications, such as early screening and treatment of autism spectrum disorder (ASD) [13, 25], social anxiety inhibition [7], and oppositional defiant disorder [5].

Although previous intensive studies [3, 20–22] have been conducted on human behavior analysis, eye contact detection and next speaker prediction remain profoundly challenging. An intuitive and direct solution to eye contact detection is to estimate the gaze direction of each participant. Previous eye gaze analysis methods can be classified into two categories (i.e., model-based and appearance-based). Model-based methods utilize geometric models of the eyeball along with the environment, and extract eye features (e.g., pupil center and corneal infrared reflections) to fit the model. However, it's difficult to accurately extract such small eye features, because the capture systems usually involve stereo camera pairs, NIR imaging devices, and thus are hard to be built and calibrate. Additionally, appearance-based approaches [15, 26] work based on the hypothesis that the participants are frontally facing the camera, which is not directly applicable for the multi-person conversation. More recently, Otsuka *et al.* [23] propose to use Convolutional Neural Networks (CNN) for eye contact detection by combining head pose, utterances and eye directions. Similarly, Fu *et al.* [6] also utilize the CNN to capture discriminative patterns from the participants' appearance features and behavioral features.

Identifying who will speak next in multi-party conversations can be based on a variety of features, such as eye gaze [12], mouth opening [11], head movement [10]. However, there are multiple candidates for the next speaker, which makes it difficult to perform next speaker prediction. For automatically and accurately predicting the next speaker in group communications, Ishii *et al.* [10] utilize a head tracker to focus the head movements of each participant. Moreover, they [11] propose to use gaze behavioral patterns to determine the next speaker in turn-changing. Different from these methods, Birmingham *et al.* [2] achieve the improved next speaker prediction by combining group-level focus of visual attention features and audio-video synchronizer models. Regarding the multimodal models, Masumura *et al.* [17] propose to utilize cross-modal representations by multimodal encoders and perform speech-based end-of-turn detection.

We believe that previous commonly used features (e.g., eye gaze, head pose, mouth opening) can be intrinsically represented by the deep visual features from images. Therefore, we adopt a computer vision-based method for group-level human behavior analysis. Specifically, we first extract and align facial regions from the raw video data. The video samples from the MPIIGroupInteraction dataset [20] and the preprocessed facial frames are shown in Fig. 1. Subsequently, we propose the Transformer-augmented convolutional neural network (TA-CNN) as the solution to both sub-challenges (i.e., eye contact detection and next speaker prediction). Experimental results on the validation set and the unpublished test set demonstrate the effectiveness of our proposed method.

2 METHODOLOGY

2.1 Task Definitions

Our proposed approach applies the TA-CNN to encoding visual features for position-independent eye contact detection and group-level next speaker prediction. The eye contact of a participant can be modelled as a discrete indication of which participant he/she is making eye contact with. Given a 10-second context window of multi-sensor data for each participant, the eye contact is detected

and inferred by the temporal context information. We model the eye contact detection as a multi-class classification problem. The position of each participant is used to transform the absolute eye contact classes to the relative ones. Therefore, the task is formulated to predict four categories (i.e., left, frontal, right and no eye contact). Similar with the eye contact detection task, the next potential speaker is also determined based on an observation window of context data. The next speaker prediction task can be defined as a multi-label classification problem, which requires the model to predict the speaking status (i.e., speaking, non-speaking) for each participant.

2.2 TA-CNN for Eye Contact Detection

Recent flourishing of Transformer-based methods on computer vision tasks [4, 16] has considerably deepened our understanding about discriminative feature representation and contextual information modeling. Therefore, we propose a simple but effective Transformer-augmented convolutional neural network (TA-CNN) for eye contact detection, which can capture and exploit the facial appearance information to enhance the performance. The proposed TA-CNN for eye contact detection is shown in Fig. 2, which mainly consists of two parts, i.e., input embedding generation, global feature learning.

Input Embedding Generation: Given an image sequence consisting of multiple facial frames as the input, we utilize a CNN backbone (e.g. ResNet18 [9]) to translate each frame into a visual feature sequence. Specifically, we first detect and align facial regions with OpenFace [1] on the last two-second window of each video, resulting in 60 frames for each sample. During the training process, each sample sequence is divided into T segments, and we uniformly sample one frame from each segment. Therefore, T frames are obtained for each video for training. In addition, we heuristically sample T frames in the mid of each segment for validation and testing.

Subsequently, a standard CNN backbone is applied for extracting high-level features for each video sample. We flatten the feature map f_0 of size $T \times H \times W \times C$ and projected it into a new feature sequence $f_1 \in \mathbb{R}^{T \times (H \times W) \times d}$, where d is the hidden dimension. We incorporate the learnable positional embeddings $e_{pos}^{space} \in 1 \times (H \times W) \times d$ with f_1 to supplement the spatial positional information, which can be formulated as

$$f_2 = f_1 + e_{pos}^{space}. \quad (1)$$

The classification token is prepended to the sequence f_2 at the temporal dimension to model the global state of the sequence, which is further used for eye contact detection. We reshape the new $f_2 \in \mathbb{R}^{(T+1) \times (H \times W) \times d}$ into $f_3 \in \mathbb{R}^{(H \times W) \times (T+1) \times d}$ and supplement the temporal information by

$$z^0 = f_3 + e_{pos}^{time}, \quad (2)$$

where $e_{pos}^{time} \in \mathbb{R}^{1 \times (T+1) \times d}$ is the temporal learnable positional embeddings. The input embedding z^0 is generated for further learning global dependencies across frames and detecting eye contact.

Global Feature Learning: The standard multi-layer Transformers are utilized to learn a discriminative representation for each frame and exploit the long-range temporal dependencies to perform eye contact detection. Each Transformer Layer consists of N blocks

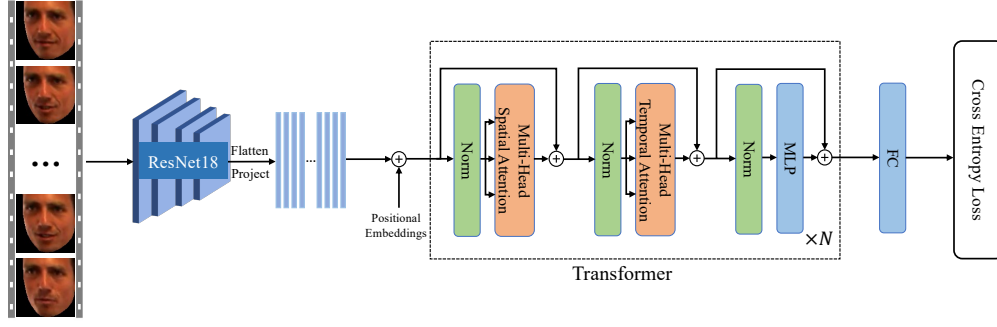


Figure 2: Overview of our method for eye contact detection. The facial frames sampled from a video sample are sent into the CNN backbone to obtain frame-level feature sequences. The Multi-layer Transformer jointly calculates the spatial attention and the temporal attention for capturing discriminative feature tokens.

and each block is composed with the multi-head spatial attention (MHSA) and the multi-head temporal attention (MHTA), which iteratively learn the contextual and discriminative spatio-temporal feature representation.

The spatial interactions are performed by reshaping the embedding z into $\mathbb{R}^{(H \times W) \times (T+1) \times d}$ (denoted by z_s). Similarly, the input to the temporal attention is reshaped into $\mathbb{R}^{(T+1) \times (H \times W) \times d}$. The spatio-temporal global feature learning at the i -th block can be defined as

$$y_s^l = \text{MHSA}(\text{LN}(z_s^l)) + z_s^l, \quad (3)$$

$$y_t^l = \text{MHTA}(\text{LN}(y_s^l)) + y_s^l, \quad (4)$$

$$z^{l+1} = \text{MLP}(\text{LN}(y_t^l)) + y_t^l, \quad (5)$$

where LN and MLP represent the layer normalization layer and the multi-layer perception layer, respectively.

The classification token $z_{(0,0)}^N$ at the beginning of z^N of the final block encodes the information from each frame and propagates it temporally across the sequence of frames. And we apply a single fully connected (FC) layer to the token $z_{(0,0)}^N$:

$$p = \text{FC}(z_{(0,0)}^N), \quad (6)$$

where p is the prediction distribution of 4 eye contact classes (i.e., left, frontal, right, no eye contact).

2.3 TA-CNN for Next Speaker Prediction

The solution to next speaker prediction is shown in Fig. 3. The main network here is consistent with that for eye contact detection. Different from the solution to eye contact detection, we concatenate the last facial frame of each participant in one group as the input. The CNN backbone is also used to generate group-level input embeddings. Similarly, we apply multi-head self attention to jointly learn the group-level feature relationships for all participants. As mentioned above, the next speaker prediction task is regarded as a multi-label classification problem. Therefore, we adopt multi classification branches to predict the speaking status for each participant. The standard cross-entropy loss is used for each branch to supervise our model in an end-to-end manner.

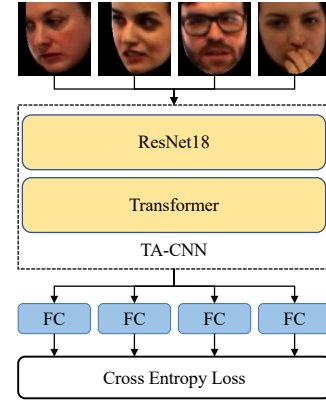


Figure 3: Overview of our method for next speaker prediction. The last frame of each participant is concatenated and processed by the TA-CNN. We apply the multi-label classification strategy for group-level next speaker prediction.

3 EXPERIMENTS

3.1 Datasets

The MULTIMEDIATE'22 Challenge utilizes the MPIIGroupInteraction dataset [20] for eye contact detection and next speaker prediction sub-challenges, which consists of 22 group interactions and 78 German-speaking participants. This dataset has been officially cropped and divided into various subsets with different volumes for training, validation and testing, which greatly minimize the overhead needed to participate in this challenge. Specifically, the dataset provides 4504 training samples, 1672 validation samples and 1848 testing samples for eye contact detection. In addition, the officially provided next speaker prediction dataset contains 7546 samples for training, 4036 samples for validation, and 5222 samples for testing.

3.2 Implementation Details

Data Preprocessing: The OpenFace toolkit [1] is used to detect and align facial regions from the raw videos. All the facial region images are resized to 112×112 pixels as the inputs to our model. For

Table 1: Accuracies for the eye contact detection sub-challenge provided by the official baseline[18] this year and our proposed methods.

Feature	Method	Acc (val)	Acc (test)
Gaze	Baseline [18]	0.61	-
Headpose	Baseline [18]	0.62	-
Gaze + Headpose	Baseline [18]	0.64	0.576
Deep Features	CNN	0.7218	-
Deep Features	TF-CNN	0.7355	0.7007
Deep Features	CNN-Seq	0.7497	0.7088
Deep Features	PA-TA-CNN	0.7420	0.6520
Deep Features	TA-CNN	0.7605	0.7148
Deep Features	TA-CNN (trainval)	-	0.7261

the next speaker prediction, we fix the number of participants in one group to 4. However, there are some samples with only three participants, which are not directly applicable for our model. We just fill the position with zero paddings to address this issue. During training, the data augmentation techniques (e.g., random crop and resize, random erasing) are adopted to avoid overfitting.

Experimental Settings: We implement our method with the PyTorch [24] platform with two NVIDIA GTX 1080Ti GPU cards. Our model is trained for 50 epochs with the batch-size of 32 and the learning rate of our network is initialized as 0.005. We use a linear learning rate warmup of 1,000 steps and cosine learning rate decay. The Adam optimizer [14] is used to optimize the whole networks. It is noted that the CNN backbone ResNet18 is pretrained on MS-Celeb-1M [8]. The number of Transformer layers N and the number of heads are empirically assigned to 2 and 8, respectively. In addition, the hidden dimension d is set to 512. For the eye contact detection task, each 2-second video clip is divided into $T = 6$ segments. $T = 6$ samples are also selected from the mid of each segment for validation and testing. For the next speaker prediction task, we only use the last frame of each video sample, which is totally different from the previous method [2]. The accuracy and the unweighted average recall (UAR) serve as the evaluation metrics for eye contact detection and next speaker prediction, respectively.

3.3 Experimental Results:

Results for Eye Contact Detection: The validation results and the test results are presented in Tab. 1. We denote the four types of model architectures as CNN, TF-CNN, CNN-Seq, TA-CNN for a pure CNN, a CNN with a two-layer Transformer, a CNN with the sequence input, our TA-CNN. The first two methods only use the last frame for eye contact detection, while the last two methods take the image sequence as the inputs. In addition, we also train four separate TA-CNNs for each participant with the same seating position, which is denoted as the position-aware TA-CNN (PA-TA-CNN). The official baseline uses a RBF-SVM on the top of gaze features and head pose features extracted with OpenFace. As we can see from Tab. 1, our baseline method CNN achieves 0.7218 in terms of accuracy on the validation set and improves the official baseline by more than 0.08, which demonstrates the computer vision-based solution is effective for this sub-challenge. With the

Table 2: Unweighted Average Recall (UAR) for the next speaker prediction sub-challenge obtained by our proposed methods. It is noted that 'S' represents 'Static' and 'D' denotes 'Dynamic', as described in the baseline paper [19]. 'Group' means the group-level features are used.

Feature	Method	UAR (val)	UAR (test)
Group S	Baseline [19]	0.60	-
Group D	Baseline [19]	0.53	-
Group (S+D)	Baseline [19]	0.60	-
Deep Features	CNN	0.5791	-
Deep Features	CNN-Seq	0.6372	0.5365
Deep Features	TA-CNN	0.6538	0.5766
Deep Features	TA-CNN (trainval)	-	0.5965

help of Transformers, our method TF-CNN further improves our baseline method by 0.0137. From the results of CNN-Seq and TA-CNN, we can draw the conclusion that the temporal dynamics also contribute to the performance improvements. Notably, we find a severe overfitting problem when training our PA-TA-CNN individually for each seating position. Finally, the TA-CNN trained on the training subset and the validation subset achieves the accuracy of 0.7261.

Results for Next Speaker Prediction: The experimental results of next Speaker prediction are shown in Tab. 2. Similar with the settings of eye contact detection, our baseline method CNN utilizes the last frame of each participant in one group and predict the speaking status individually. When trained with multiple facial regions from the same group, the CNN-Seq improves the baseline by 0.0581, which indicates that the group-level information is essential to make accurate predictions. Our group-level features-based models (i.e., CNN-Seq, TA-CNN) improves the baseline results [19] by a wide margin. Our TA-CNN can better exploit and capture the discriminative tokens from the group-level features, resulting in the UAR of 0.6538 and 0.5766 on the validate set and the test set, respectively. The TA-CNN achieves 0.5965 on the unpublished test set by training with both validation and test sets.

4 CONCLUSION

In this paper, we present our solutions to the eye contact detection and the next speaker prediction sub-challenges of MULTIMEDIATE'22. A unified and effective network TA-CNN is proposed for human behavior analysis in multi-person conversations. Specifically, we adopt a computer vision-based strategy by extracting facial regions from the video data and sending them into deep neural networks. The proposed TA-CNN network can simply change the input settings for both tasks and learn the temporal or group-level features to achieve satisfactory performances. The experimental results on the validation and the unpublished test sets demonstrate that our method has good generalization ability for both sub-challenges.

5 ACKNOWLEDGMENTS

This work was supported by the National Natural Science Fund of China (62171183) and by Hunan Provincial Natural Science Foundation of China (2022JJ20017).

REFERENCES

- [1] Tadas Baltrušaitis, Amir Zadeh, Yao Chong Lim, and Louis-Philippe Morency. 2018. Openface 2.0: Facial behavior analysis toolkit. In *IEEE International Conference on Automatic Face & Gesture Recognition (FG 2018)*. IEEE, 59–66.
- [2] Chris Birmingham, Kalin Stefanov, and Maja J Mataric. 2021. Group-Level Focus of Visual Attention for Improved Next Speaker Prediction. In *Proceedings of the 29th ACM International Conference on Multimedia*. 4838–4842.
- [3] Michael Dietz, Daniel Schork, and Elisabeth André. 2016. Exploring eye-tracking-based detection of visual search for elderly people. In *2016 12th International Conference on Intelligent Environments (IE)*. IEEE, 151–154.
- [4] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xi-aohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, et al. 2020. An image is worth 16x16 words: Transformers for image recognition at scale. *arXiv preprint arXiv:2010.11929* (2020).
- [5] Lourdes Ezpeleta, Roser Granero, Núria De La Osa, and Josep M Domènech. 2015. Clinical characteristics of preschool children with oppositional defiant disorder and callous-unemotional traits. *PLoS one* 10, 9 (2015), e0139346.
- [6] Eugene Yujun Fu and Michael W Ngai. 2021. Using Motion Histories for Eye Contact Detection in Multiperson Group Conversations. In *Proceedings of the 29th ACM International Conference on Multimedia*. 4873–4877.
- [7] Xiaoxue Fu, Eric E Nelson, Marcela Borge, Kristin A Buss, and Koral Pérez-Edgar. 2019. Stationary and ambulatory attention patterns are differentially associated with early temperamental risk for socioemotional problems: Preliminary evidence from a multimodal eye-tracking investigation. *Development and Psychopathology* 31, 3 (2019), 971–988.
- [8] Yandong Guo, Lei Zhang, Yuxiao Hu, Xiaodong He, and Jianfeng Gao. 2016. Ms-celeb-1m: A dataset and benchmark for large-scale face recognition. In *European Conference on Computer Vision*. Springer, 87–102.
- [9] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. 2016. Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. 770–778.
- [10] Ryo Ishii, Shiro Kumano, and Kazuhiro Otsuka. 2015. Predicting next speaker based on head movement in multi-party meetings. In *2015 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2319–2323.
- [11] Ryo Ishii, Kazuhiro Otsuka, Shiro Kumano, Ryuichiro Higashinaka, and Junji Tomita. 2019. Prediction of who will be next speaker and when using mouth-opening pattern in multi-party conversation. *Multimodal Technologies and Interaction* 3, 4 (2019), 70.
- [12] Ryo Ishii, Kazuhiro Otsuka, Shiro Kumano, and Junji Yamato. 2016. Prediction of who will be the next speaker and when using gaze behavior in multiparty meetings. *ACM Transactions on Interactive Intelligent Systems (TIIS)* 6, 1 (2016), 1–31.
- [13] Shomik Jain, Balasubramanian Thiagarajan, Zhonghao Shi, Caitlyn Clabaugh, and Maja J Mataric. 2020. Modeling engagement in long-term, in-home socially assistive robot interventions for children with autism spectrum disorders. *Science Robotics* 5, 39 (2020), eaaz3791.
- [14] Diederik P Kingma and Jimmy Ba. 2014. Adam: A method for stochastic optimization. *arXiv preprint arXiv:1412.6980* (2014).
- [15] Feng Lu, Yusuke Sugano, Takahiro Okabe, and Yoichi Sato. 2014. Adaptive linear regression for appearance-based gaze estimation. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 36, 10 (2014), 2033–2046.
- [16] Fuyan Ma, Bin Sun, and Shutao Li. 2021. Facial Expression Recognition with Visual Transformers and Attentional Selective Fusion. *IEEE Transactions on Affective Computing* (2021). doi: 10.1109/TAFFC.2021.3122146.
- [17] Ryo Masumura, Mana Ithori, Tomohiro Tanaka, Atsushi Ando, Ryo Ishii, Takanobu Oba, and Ryuichiro Higashinaka. 2019. Improving speech-based end-of-turn detection via cross-modal representation learning with punctuated text data. In *2019 IEEE Automatic Speech Recognition and Understanding Workshop (ASRU)*. IEEE, 1062–1069.
- [18] Philipp Müller, Michael Dietz, Dominik Schiller, Dominike Thomas, Hali Lindsay, Patrick Gebhard, Elisabeth André, and Andreas Bulling. 2022. MultiMediate '22: Backchannel Detection and Agreement Estimation in Group Interactions. In *Proceedings of the 30th ACM International Conference on Multimedia*. ACM New York, NY, USA, 6 pages. <https://doi.org/10.1145/3503161.3551589>
- [19] Philipp Müller, Michael Dietz, Dominik Schiller, Dominike Thomas, Guanhua Zhang, Patrick Gebhard, Elisabeth André, and Andreas Bulling. 2021. MultiMediate: Multi-modal Group Behaviour Analysis for Artificial Mediation. In *Proceedings of the 29th ACM International Conference on Multimedia*. 4878–4882.
- [20] Philipp Müller, Michael Xuelin Huang, and Andreas Bulling. 2018. Detecting low rapport during natural interactions in small groups from non-verbal behaviour. In *23rd International Conference on Intelligent User Interfaces*. 153–164.
- [21] Philipp Müller, Michael Xuelin Huang, Xucong Zhang, and Andreas Bulling. 2018. Robust eye contact detection in natural multi-person interactions using gaze and speaking behaviour. In *Proceedings of the 2018 ACM Symposium on Eye Tracking Research & Applications*. 1–10.
- [22] Philipp Müller, Ekta Sood, and Andreas Bulling. 2020. Anticipating averted gaze in dyadic interactions. In *ACM Symposium on Eye Tracking Research and Applications*. 1–10.
- [23] Kazuhiro Otsuka, Keisuke Kasuga, and Martina Köhler. 2018. Estimating visual focus of attention in multiparty meetings using deep convolutional neural networks. In *Proceedings of the 20th ACM International Conference on Multimodal Interaction*. 191–199.
- [24] Adam Paszke, Sam Gross, Francisco Massa, Adam Lerer, James Bradbury, Gregory Chanan, Trevor Killeen, Zeming Lin, Natalia Gimelshein, Luca Antiga, et al. 2019. Pytorch: An imperative style, high-performance deep learning library. *arXiv preprint arXiv:1912.01703* (2019).
- [25] Brian Scassellati, Laura Boccanfuso, Chien-Ming Huang, Marilena Mademtzi, Meiyang Qin, Nicole Salomons, Pamela Ventola, and Frederick Shic. 2018. Improving social skills in children with ASD using a long-term, in-home social robot. *Science Robotics* 3, 21 (2018), eaat7544.
- [26] Xucong Zhang, Yusuke Sugano, Mario Fritz, and Andreas Bulling. 2015. Appearance-based gaze estimation in the wild. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. 4511–4520.